

DeshIT Hospital Management System

Complete System Modules & Submodules Documentation (Exact Navbar Order)

1. System Architecture & Navigation Overview

This document presents the complete hierarchy of all 20 Navbar Modules and their exact Entry & Record submodules as structured in the application navigation header (from left to right).

Technology Stack Breakdown:

- Frontend UI Layer: React 18, Vite, React Router DOM v7 (src/features/*).
- Backend API Layer: Node.js, Express.js (src/modules/* controllers, services & routes).
- Database Storage Layer: MySQL (hospital_management) with 37 Relational Tables & Sequelize ORM.
- Security Protocol: JWT Access Tokens, Refresh Tokens, Joi Validation, RBAC Guards & Audit Logging.

2. All 20 Modules & Submodules (Exact Top Navbar Order)

1. Options (System Master Settings)

Frontend Path: src/features/options | Backend Path: src/modules/settings

Database Tables: settings, master_options

Entry Submodules (Data Input Forms):

- Master Options: Configures symptom types, symptom heads, charge categories & tax rates.
- System Settings: Hospital company profile (Name, Logo, Phone, Address, Currency, VAT).

Record Submodules (Audit Records & Logs):

- Tariff Charges: Master price tariff list for all hospital services & diagnostic charges.

2. Quick Options (Quick Shortcuts)

Frontend Path: src/features/quick-options | Backend Path: src/modules/settings

Database Tables: patients, opd_visits, lab_orders, pharmacy_sales

Entry Submodules (Data Input Forms):

- Quick Patient Registration: Fast-track patient intake form.
- Quick OPD Consultation Ticket: Instant ticket issuance for outpatient visits.
- Quick Lab Test Booking: Fast diagnostic pathology test ticket creation.
- Quick Pharmacy Sale: Rapid POS counter medicine sale.

Record Submodules (Audit Records & Logs):

- Today's Summary: Real-time quick access transaction summary.

3. Dashboard (Executive Overview)

Frontend Path: src/features/dashboard | Backend Path: src/modules/reports

Database Tables: patients, opd_visits, admissions, beds, medicine_sales, invoices

Entry Submodules (Data Input Forms):

- Live Metrics Overview: Displays total patient count, today's OPD visits & active IPD bed occupancy.
- Financial Revenue Widgets: Real-time graphs for daily cash collections, pharmacy sales & lab billing.

Record Submodules (Audit Records & Logs):

- Recent Activity Audit Log: System audit log of recent transactions & staff actions.

4. Modules (Central Navigation Grid)

Frontend Path: src/features/modules | Backend Path: src/modules/auth

Database Tables: users, audit_logs

Entry Submodules (Data Input Forms):

- Module Access Grid: Central visual dashboard displaying all hospital feature modules.

Record Submodules (Audit Records & Logs):

- Role Access Summary: Shows module permissions assigned to current user role.

5. Pathology (Pathology Laboratory)

Frontend Path: src/features/pathology | Backend Path: src/modules/laboratory

Database Tables: lab_tests, lab_orders, lab_order_items, master_options

Entry Submodules (Data Input Forms):

- Pathology Bill Entry: Issues patient test tickets & specimen collection slips.
- Pathology Test Entry: Configures lab tests (CBC, RBS, LFT, KFT) and price rates.
- Pathology Test Category Entry: Categorizes tests (Hematology, Biochemistry, Serology).
- Pathology Test Unit Entry: Defines test measurement units (g/dL, mg/dL, /cu.mm).
- Pathology Parameter Entry: Sets reference normal ranges for test findings.

Record Submodules (Audit Records & Logs):

- Pathology Bill Record: History of patient pathology bill tickets and lab reports.
- Pathology Test List: Master catalog of configured lab diagnostic tests.

6. Radiology (Diagnostic Imaging)

Frontend Path: `src/features/radiology` | Backend Path: `src/modules/radiology`

Database Tables: `radiology_tests`, `radiology_orders`

Entry Submodules (Data Input Forms):

- Radiology Bill Entry: Books imaging test tickets (X-Ray, USG, CT Scan, MRI, ECG).
- Radiology Test Entry: Configures imaging modalities and pricing tariffs.
- Radiology Test Category Entry: Categorizes imaging types (X-Ray, Ultrasound, CT, MRI).
- Radiology Test Unit Entry: Defines measurement parameter units.
- Radiology Parameter Entry: Sets up radiologist examination templates.

Record Submodules (Audit Records & Logs):

- Radiology Bill Record: History of radiology bill orders & patient imaging reports.
- Radiology Test List: Directory of diagnostic radiology imaging tests.

7. Appointment (Doctor Appointments)

Frontend Path: `src/features/appointment` | Backend Path: `src/modules/appointments`

Database Tables: `appointments`, `patients`, `doctors`, `departments`

Entry Submodules (Data Input Forms):

- Appointment Entry: Books doctor appointment slot & calculates consultation fee.
- Appointment Priority Entry: Sets priority levels (Emergency, Normal, VIP).
- Appointment Shift Entry: Configures morning/evening consultation shifts.
- Appointment Slots Page: Manages doctor time slot availability.

Record Submodules (Audit Records & Logs):

- Appointment Bill Record: History of booked appointments, queue serials & status.

8. OPD (Outpatient Department)

Frontend Path: `src/features/opd` | Backend Path: `src/modules/opd`

Database Tables: `opd_visits`, `appointments`, `patients`, `doctors`

Entry Submodules (Data Input Forms):

- OPD Bill Entry: Issues outpatient consultation ticket, records symptoms & vitals.
- OPD Symptoms Manage: Configures symptom types & chief complaint categories.

Record Submodules (Audit Records & Logs):

- OPD Bill Record: Audit list of outpatient visits, consultation fees & doctor notes.

9. IPD (Inpatient & Admissions)

Frontend Path: `src/features/ipd` | Backend Path: `src/modules/ipd`

Database Tables: `admissions`, `patients`, `doctors`, `beds`, `wards`

Entry Submodules (Data Input Forms):

- IPD Bill Entry: Admits patient to hospital bed (ADM-XXX), sets diagnosis & advance paid.
- IPD Symptoms Manage: Manages inpatient admission symptom categories.

Record Submodules (Audit Records & Logs):

- IPD Bill Record: Inpatient directory, stay duration, daily room charges & discharge summaries.

10. Ambulance (Emergency Transport)

Frontend Path: `src/features/ambulance` | Backend Path: `src/modules/ambulance`

Database Tables: `ambulances`, `ambulance_trips`

Entry Submodules (Data Input Forms):

- Call Ambulance Entry: Dispatches emergency ambulance trip, records pickup/dropoff & fare.
- Ambulance Entry: Registers ambulance fleet vehicles (ICU/Standard) & driver details.

Record Submodules (Audit Records & Logs):

- Call Ambulance Record: History of ambulance trip dispatches, distance km & fare invoices.

11. Blood (Blood Bank Unit)

Frontend Path: `src/features/blood` | Backend Path: `src/modules/blood-bank`

Database Tables: `blood_donors`, `blood_bags`, `blood_issues`

Entry Submodules (Data Input Forms):

- Blood Donate Entry: Registers blood donation from donor.
- Blood Issue Entry: Requisitions blood bag for emergency or IPD patient.
- Blood Component Issue Entry: Issues specific blood components (RBC, Plasma, Platelets).
- Blood Component Separation Entry: Records separation of whole blood into components.
- Blood Donor Entry: Enrolls new blood donor profiles and contact info.
- Blood Group Entry: Configures blood group categories (A+, B+, O+, AB+).
- Blood Unit Entry: Configures blood measurement units.
- Blood Component Entry: Registers component types (Plasma, Platelets, Cryo).

Record Submodules (Audit Records & Logs):

- Blood Stock: Real-time inventory of available blood bags by group.
- Blood Donate Record: Logs of donor blood donations.
- Blood Issue Record: Audit list of blood bag transfusions & issues.
- Blood Component Issue Record: Logs of blood component issues.
- Blood Component Separation Record: History of blood component separations.

12. Pharmacy (POS & Inventory)

Frontend Path: `src/features/pharmacy` | Backend Path: `src/modules/pharmacy`

Database Tables: medicines, medicine_sales, medicine_sale_items

Entry Submodules (Data Input Forms):

- Pharmacy Sales: POS Counter sales memo for dispensing medicine to patients.
- Pharmacy Purchase: Records bulk medicine inventory purchase from suppliers.
- Pharmacy Sales Return: Processes patient medicine sales returns & refunds.
- Pharmacy Purchase Return: Handles returns of expired/damaged stock to suppliers.
- Pharmacy Supplier: Manages pharmaceutical supplier company directory.
- Medicine Batch Stock: Tracks batch numbers, manufacturing & expiry dates.

Record Submodules (Audit Records & Logs):

- Pharmacy Sales Record: Audit list of pharmacy sales memo transactions.
- Pharmacy Purchase Record: Purchase order history from suppliers.
- Pharmacy Sales Return Record: Logs of medicine sales return memos.
- Pharmacy Purchase Return Record: Logs of supplier purchase return memos.
- Medicine Stock Report: Inventory stock levels & reorder alert status.

13. Medicine (Medicine Catalog)

Frontend Path: `src/features/medicine` | Backend Path: `src/modules/pharmacy`

Database Tables: medicines

Entry Submodules (Data Input Forms):

- Medicine Entry: Registers new medicine items, generic names, prices & stock.
- Medicine Category Entry: Configures medicine categories (Tablet, Capsule, Syrup).
- Medicine Company Entry: Configures pharmaceutical manufacturing companies.
- Medicine Group Entry: Configures therapeutic generic groups.
- Medicine Unit Entry: Sets up dispensing unit metrics (Pcs, Box, Strip).

Record Submodules (Audit Records & Logs):

- Medicine List: Directory of all configured medicines and stock status.

14. Finance (Accounts & Financials)

Frontend Path: `src/features/finance` | Backend Path: `src/modules/billing`

Database Tables: expenses, expense_categories, invoices, payments, master_options

Entry Submodules (Data Input Forms):

- Expense Entry: Enters daily operational hospital expense vouchers and payment methods.
- Income Entry: Registers non-billing hospital incomes and direct cash collections.
- Contra Entry: Records internal bank-to-cash or cash-to-bank fund transfers.
- Expense Head Entry: Configures expense category heads (Utility, Supplies, Repairs).
- Income Head Entry: Configures revenue income heads and accounts.
- Tax Rate: Sets up VAT & Tax percentage rates (0%, 5% VAT).
- Account Entry: Manages cash and bank payment account ledgers.

Record Submodules (Audit Records & Logs):

- Contra Record: Logs history of all contra fund transfers.
- Expense Record: Audit list of daily operational expenses.
- Income Record: Logs history of all income collection vouchers.
- Patient Due Collection List: Tracks patient due bills and pending payment collections.

15. Reports (Executive Analytics)

Frontend Path: `src/features/reports` | Backend Path: `src/modules/reports`

Database Tables: invoices, payments, expenses, opd_visits, lab_orders, radiology_orders, medicine_sales

Entry Submodules (Data Input Forms):

- Daily Statement Report: Sectional revenue breakdown (Pathology, OPD, IPD, Pharmacy, Radiology).
- Daily Ledger Report: Daily cashbook and financial transaction log.
- Patient Ledger Report: Individual patient debit/credit account ledger.
- Patient Balance Report: Summary of patient outstanding due balances.
- Account Ledger Report: Cash & bank account transaction ledger.
- Account Balance Report: Financial balance summary across bank accounts.
- Pharmacy Stock Report: Pharmacy inventory valuation & stock summary.
- Medicine Batch-wise Stock Report: Batch number & expiry date stock audit.
- Supplier Balance Report: Vendor and pharmaceutical supplier due balances.
- Supplier Ledger Report: Individual supplier transaction history ledger.

Record Submodules (Audit Records & Logs):

- OPD Bill Record Report: Outpatient billing audit report.
- IPD Bill Record Report: Inpatient billing audit report.
- Pathology Bill Record Report: Lab diagnostic billing audit report.
- Radiology Bill Record Report: Imaging billing audit report.
- Call Ambulance Record Report: Ambulance trip financial report.
- Referral Bill Record Report: Referral commission payout report.

16. BED (Bed & Ward Allocation)

Frontend Path: src/features/bed | Backend Path: src/modules/beds

Database Tables: beds, wards

Entry Submodules (Data Input Forms):

- Bed Entry: Configures bed numbers, ward mapping & daily room rates.
- Bed Management: Grid layout tool for bed maintenance and status.

Record Submodules (Audit Records & Logs):

- Bed Availability: Real-time live occupancy view of available/occupied beds.

17. Patient (Patient Registry)

Frontend Path: src/features/patient | Backend Path: src/modules/patients

Database Tables: patients

Entry Submodules (Data Input Forms):

- Patient Entry: Registers new patient profiles and generates PAT-ID.

Record Submodules (Audit Records & Logs):

- Patient List: Searchable directory of registered hospital patients.

18. Doctor (Doctors & Schedules)

Frontend Path: src/features/doctor | Backend Path: src/modules/doctors

Database Tables: doctors, departments, users

Entry Submodules (Data Input Forms):

- Doctor Entry: Registers doctor profile, consultation fee & qualifications.
- Doctor Department: Maps doctor to medical department.
- Doctor Specialization: Configures medical specialties.

Record Submodules (Audit Records & Logs):

- Doctor List: Directory of hospital doctors and chamber schedules.

19. Referral (Referral Management)

Frontend Path: src/features/referral | Backend Path: src/modules/referrals

Database Tables: referrals, doctors, patients

Entry Submodules (Data Input Forms):

- Referral Bill Entry: Records patient referral bill and commission.
- Referral Person Entry: Registers external referral doctors & clinics.

Record Submodules (Audit Records & Logs):

- Referral Bill Record: History of patient referrals & commission status.
- Referral Person List: Directory of referral persons and external facilities.

20. HR & Payroll (Human Resources)

Frontend Path: src/features/hr-payroll | Backend Path: src/modules/hr

Database Tables: employees, attendance, payrolls, departments

Entry Submodules (Data Input Forms):

- Employee Entry: Registers hospital staff employees, designations & basic salary.
- Department Entry: Configures HR organizational departments.
- Employee Salary Payment Entry: Processes monthly salary payments & allowances.
- Salary Sheet Page: Generates monthly hospital staff salary sheet.

Record Submodules (Audit Records & Logs):

- Employee List: Directory of active hospital staff and designation details.

System Status: 100% Fully Backend-Driven & Database Verified

Database: MySQL (hospital_management) | Total Verified Tables: 37 | All APIs Active on Port 5000